A Geometric Approach to Firm-Type Factor Models

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1 Introduction

A vast asset pricing literature has long relied on firm characteristics to explain the cross-section of expected returns. The traditional approach suggests that sorting assets into discrete portfolios based on observable attributes effectively captures exposure to underlying, systematic risk factors. Recent advancements in latent factor modeling have demonstrated that the entire high-dimensional panel of firm characteristics can be compressed to pinpoint these compensated aggregate risk exposures more precisely. In these frameworks, a firm's cross-sectional profile is represented by a high-dimensional vector of characteristics, and dimension reduction techniques are used to identify a lower-dimensional set of latent factors.

Standard approaches to latent factor modeling and characteristic-based portfolio sorting implicitly rely on linear mappings or Euclidean distance metrics to define firm similarity and factor exposures. For instance, observable factor models in the tradition of Fama and French (1993, 2015), as well as latent factor models like principal component analysis (PCA) and instrumented PCA (IPCA; Kelly et al., 2019), conventionally group assets by minimizing squared Euclidean distances or relying on linear characteristic exposures. This mechanical assumption treats both the magnitude and the direction of the characteristic vector $z_{i,t-1} \in \mathbb{R}^P$ as equally informative when determining a firm's underlying type.

There is, however, little theoretical or intuitive justification for relying on Euclidean magnitude when comparing cross-sectional firm profiles. To the contrary, standard distance metrics often fail to capture the purely directional, scale-invariant nature of a firm's fundamental "type." For example, a firm with twice the market capitalization and twice the book value of another firm may share the exact same fundamental risk profile. Standard Euclidean metrics penalize this size difference, treating the firms as distinct, whereas a directional metric recognizes they represent the same proportional archetype. By penalizing differences in vector magnitude rather than orientation, Euclidean-based models risk conflating fundamentally distinct firm profiles or separating firms that share identical proportional risk exposures. Consequently, this geometric mismatch leads to noisy latent factor exposures and suboptimal risk premia estimation.

To resolve this, we propose that a firm's latent risk profile is best defined by the relative proportions of its characteristics rather than their absolute magnitudes. We generalize the construction of characteristic-based factor models using concepts from spherical geometry. Specifically, we retrofit the workhorse unsupervised clustering device—K-means—by projecting the high-dimensional characteristic space onto a unit hypersphere. By mapping characteristics to a directional representation $u_{i,t-1} \in \mathbb{S}^{P-1}$, we isolate the purely proportional profile of each firm. We then employ Spherical K-Means to identify latent firm types by maximizing intra-cluster cosine similarity rather than minimizing Euclidean distance.

In this paper, we introduce a new geometric firm-type factor model which utilizes this spherical manifold to determine conditional risk exposures. Our approach uses distance-to-centroid normalization to relax hard clustering assignments into time-varying, soft-membership weights $\pi_{i,m,t-1}$. At the same time, we make economically guided choices in structuring this mapping, transforming these soft scores into implementable, self-financing, dollar-neutral factor-mimicking portfolios. Ultimately, ours is a geometrically-derived conditional asset pricing model, where nonlinear factor exposures manifest through a firm's cosine proximity to latent archetypes on the unit sphere.

Our empirical analysis of the panel of individual US stock returns shows that our geometric firm-type factor model dominates standard Euclidean-based clustering methods and traditional linear factor models. We conduct our comparisons on a purely out-of-sample basis, benchmarking our spherical approach against the Fama-French five-factor model, standard PCA, and IPCA.

[We find that the out-of-sample pricing errors from our spherical model are far smaller and generally insignificant compared to these leading linear alternatives.]

We also compare the relative performance of these models in economic terms. By forming tradable factor-combination portfolios using a constrained mean-variance optimization across our derived firm-type factors, we evaluate the implementable value of the spherical geometry approach.

[The optimized portfolio utilizing our geometric firm-type factors earns an annualized out-of-sample Sharpe Ratio of X.XX, significantly outperforming the Y.YY Sharpe ratio of the IPCA and Fama-French baselines.]

In summary, acknowledging the directional geometry of firm characteristics yields a conditional factor model that outperforms its leading competitors by a wide margin.

2 Pipeline

2.1 Data

Please reference this Dropbox file: `https://ucla.box.com/s/t9lmyz7lwiamff70peo0bqj1ns2vm0jj`
This has all 94 monthly firm characteristics and their respective descriptions.

2.2 Normalization

Let $z_{i,t-1} \in \mathbb{R}^P$ denote the vector of lagged characteristics for asset i observed at time $t-1$. Characteristics are standardized cross-sectionally and mapped to a directional representation via

$$u_{i,t-1} = \frac{z_{i,t-1}}{\|z_{i,t-1}\|_2}, \tag{1}$$

so that $u_{i,t-1} \in \mathbb{S}^{P-1}$.

Similarity between firms is measured using cosine similarity,

$$\mathrm{sim}(i, j) = u_{i,t-1}^\top u_{j,t-1}.$$

Given the directional embeddings $u_{i,t-1}$ defined in Equation (1), we define the *cosine distance* between assets i and j as:

$$d(i, j) = 1 - \mathrm{sim}(i, j) = 1 - u_{i,t-1}^\top u_{j,t-1} \tag{2}$$

The corresponding $N \times N$ cosine distance matrix $\mathbf{D}_{t-1}$ is constructed such that its elements are:

$$[\mathbf{D}_{t-1}]_{ij} = 1 - \frac{z_{i,t-1}^\top z_{j,t-1}}{\|z_{i,t-1}\|_2 \|z_{j,t-1}\|_2} \tag{3}$$

2.3 Clustering

To identify the M latent firm types from the directional embeddings $u_{i,t-1} \in \mathbb{S}^{P-1}$, we employ two distinct clustering paradigms: a partition-based approach optimized for spherical geometry, and a density-based approach that utilizes the cosine distance matrix $\mathbf{D}_{t-1}$.

2.3.1 Spherical K-Means

The first approach is a variant of the K-means algorithm specifically adapted for data on a unit hypersphere. Unlike standard K-means which minimizes squared Euclidean distances, Spherical K-Means seeks to maximize the intra-cluster cosine similarity. For a fixed number of M clusters, the algorithm minimizes the objective function:

$$\mathcal{J}_{sph} = \sum_{m=1}^{M} \sum_{i \in \mathcal{C}_m} (1 - u_{i,t-1}^\top \mu_m) \tag{4}$$

where μ_m is the centroid of cluster m, constrained such that $\|\mu_m\|_2 = 1$. The centroids represent the "average" directional characteristic profile for each firm type. This yields the hard membership assignments which can be relaxed into the soft weights $\pi_{i,m,t-1}$ defined in Equation (7) via distance-to-centroid normalization.

2.3.2 Hierarchical Density-Based Clustering (HDBSCAN)

The second approach utilizes HDBSCAN (Hierarchical Density-Based Spatial Clustering of Applications with Noise), which does not require a pre-specified number of clusters M and can identify clusters of arbitrary shapes on the manifold.

We define the clustering over the metric space $(\mathcal{X}, d)$, where d is the cosine distance defined by the matrix $\mathbf{D}_{t-1}$. HDBSCAN transforms this space into a mutual reachability distance to filter low-density noise:

$$d_{mreab}(i, j) = \max\{\mathrm{core}_k(i), \mathrm{core}_k(j), d(i, j)\} \tag{5}$$

where $\mathrm{core}_k(i)$ is the cosine distance to the k-th nearest neighbor of asset i. By constructing a minimum spanning tree over this distance, the algorithm extracts a stable hierarchy of firm types. This is particularly advantageous for identifying "niche" firm types that may be dense in characteristic space but small in total market representation.

Note that we will only use HBDSCAN for cluster validation (i.e. this will come up again in the validation section), but I wanted to mention it now.

2.3.3 Clustering Output and Firm-Type Mapping

The primary output of the clustering stage is the construction of the time-varying membership matrix $\Pi_{t-1} \in \mathbb{R}^{N \times M}$. For each asset i, the soft-clustering assignment $\pi_{i,m,t-1}$ represents the weight of asset i in firm-type m at time $t-1$.

In the case of Spherical K-Means, these weights are derived from the cosine distance to the centroids μ_m:

$$\pi_{i,m,t-1} = \frac{\exp(-\tau \cdot d(i, \mu_m))}{\sum_{j=1}^{M} \exp(-\tau \cdot d(i, \mu_j))} \tag{6}$$

where τ is a temperature parameter controlling the "softness" of the assignment. This mapping ensures that each asset's return can be decomposed into a linear combination of M latent firm-type factors, as detailed in the Pricing Models in Section 2.4.

2.4 Pricing Models

Throughout, let $r_{i,t}$ denote the *excess* return of asset i over $[t-1, t]$ (e.g., monthly), and let $\Pi_{t-1} = [\pi_{i,m,t-1}]_{i=1..N, m=1..M}$ denote the soft firm-type membership matrix produced by equation 7. All characteristics, memberships, and any quantities used to form weights are computed using information available at $t-1$ (no look-ahead).

Recall that firms are now grouped into M latent types using a clustering algorithm defined on the unit sphere. This yields soft cluster membership weights

$$\pi_{i,m,t-1} \in [0, 1], \qquad \sum_{m=1}^{M} \pi_{i,m,t-1} = 1, \tag{7}$$

where $\pi_{i,m,t-1}$ denotes the exposure of firm i to firm type m at time $t-1$.

2.4.1 Specification A: Firm types as priced characteristics (Fama–MacBeth)

Model. In the first specification, firm-type exposures are treated as *priced characteristics*. At each time t, we estimate the cross-sectional regression

$$r_{i,t} = \alpha_t + \sum_{m=1}^{M} \pi_{i,m,t-1} \lambda_{m,t} + \varepsilon_{i,t}, \qquad i = 1, \ldots, N, \tag{8}$$

where $\lambda_{m,t}$ is the period-t price of risk associated with firm type m. This is the expanded methodology for the "forecastable risk-premia" example I briefly went over last Thursday.

Stacking across assets, let $r_t \in \mathbb{R}^N$ and $\mathbf{1} \in \mathbb{R}^N$. Define the regressor matrix

$$X_{t-1} := [\mathbf{1}, \Pi_{t-1}] \in \mathbb{R}^{N \times (M+1)}, \qquad \theta_t := (\alpha_t, \lambda_{1,t}, \ldots, \lambda_{M,t})^\top.$$

Then the first-pass estimate is

$$\hat{\theta}_t = \arg\min_{\theta \in \mathbb{R}^{M+1}} \sum_{i=1}^{N} \omega_{i,t} \left(r_{i,t} - X_{t-1}(i, :)\theta \right)^2, \tag{9}$$

where $\omega_{i,t} \geq 0$ are cross-sectional weights (e.g., equal-weight $\omega_{i,t} = 1$ or value-weight $\omega_{i,t} \propto \text{mktcap}_{i,t-1}$). The estimated prices of risk are $\hat{\lambda}_{m,t}$.

Risk premia estimation and inference. Risk premia are estimated as time-series averages:

$$\hat{\lambda}_m = \frac{1}{T} \sum_{t=1}^{T} \hat{\lambda}_{m,t}, \tag{10}$$

with inference conducted using heteroskedasticity- and autocorrelation-consistent (HAC) standard errors computed from the time series $\{\hat{\lambda}_{m,t}\}_{t=1}^{T}$.

From priced firm types to a tradable return. Equation (8) yields *conditional expected return forecasts* at time $t-1$:

$$\hat{\mu}_{i,t|t-1} := \widehat{\mathbb{E}}[r_{i,t} \mid \mathcal{F}_{t-1}] = \hat{\alpha}_t + \sum_{m=1}^{M} \pi_{i,m,t-1} \, \hat{\lambda}_{m,t}. \tag{11}$$

In implementation, we use an expanding or rolling window to estimate $\hat{\lambda}_{m,t}$ using data up through $t-1$, and then form $\hat{\mu}_{i,t|t-1}$ for portfolio construction.

Portfolio optimization (constraints are explicit and enforced). Let $w_{t-1} \in \mathbb{R}^N$ denote the portfolio weights held over $[t-1, t]$. We define a constrained mean–variance problem using the forecast vector $\hat{\mu}_{t|t-1} = (\hat{\mu}_{1,t|t-1}, \ldots, \hat{\mu}_{N,t|t-1})^\top$ and a risk model $\hat{\Sigma}_{t-1}$ (e.g., sample covariance of returns, shrinkage covariance, or factor-based covariance).

$$\begin{aligned}
w^\star_{t-1} \; \in \; \arg\max_{w \in \mathbb{R}^N} \quad & w^\top \hat{\mu}_{t|t-1} \; - \; \frac{\gamma}{2} w^\top \hat{\Sigma}_{t-1} \, w \; - \; \kappa \, \|w - w^\star_{t-2}\|_1 \\
\text{s.t.} \quad & \mathbf{1}^\top w = 0 \qquad \text{(self-financing / dollar-neutral)} \\
& \|w\|_1 \leq L \qquad \text{(gross leverage cap)} \\
& \|w\|_\infty \leq w_{\max} \qquad \text{(single-name cap)} \\
& w \in \mathcal{W}_{t-1} \qquad \text{(tradability filters: liquidity, price, borrow, etc.)} \\
& A_{t-1} w = 0 \qquad \text{(optional neutrality constraints; e.g. market, industry, size).}
\end{aligned} \tag{12}$$

Here $\gamma > 0$ is the risk-aversion parameter, $\kappa \geq 0$ penalizes turnover via an ℓ_1 transaction-cost proxy, L bounds gross leverage, $w_{\max}$ bounds single-name concentration, and $\mathcal{W}_{t-1}$ denotes the feasible set after removing assets that fail implementation constraints at $t-1$. The linear constraints $A_{t-1} w = 0$ are used when we require explicit neutrality (examples include: market beta neutrality using an estimated beta vector, industry neutrality using a one-hot sector matrix, or characteristic neutrality to reduce unintended exposures).

Final return output (Specification A). Given the optimized weights $w^\star_{t-1}$, the realized strategy return over $[t-1,t]$ is

$$R_t^{(A)} = \sum_{i=1}^{N} w^\star_{i,t-1}\, r_{i,t} \;-\; \mathrm{TC}_t, \tag{13}$$

where TC_t is the realized transaction cost model used in backtesting and validation (e.g., proportional to turnover, bid–ask spread proxies, and/or market impact). Realistically this value will often be negligible for us... at least until we get our ISDA. aq All constraints in (12) are enforced each period.

2.4.2 Specification B: Firm-type portfolios as factor returns

Tradable firm-type factor construction. Alternatively, firm types may be represented as tradable long–short factor portfolios. For each firm type m, define a *score* at $t-1$ using the membership weights:

$$s^{(m)}_{i,t-1} := \pi_{i,m,t-1}.$$

We transform scores into a self-financing portfolio via cross-sectional demeaning and normalization. Let

$$\tilde{w}^{(m)}_{i,t-1} = s^{(m)}_{i,t-1} - \frac{1}{N}\sum_{j=1}^{N} s^{(m)}_{j,t-1}, \qquad w^{(m)}_{i,t-1} = \frac{\tilde{w}^{(m)}_{i,t-1}}{\sum_{j=1}^{N}\left|\tilde{w}^{(m)}_{j,t-1}\right|}, \tag{14}$$

so that $\sum_i w^{(m)}_{i,t-1} = 0$ and $\|w^{(m)}_{\cdot,t-1}\|_1 = 1$. This yields an implementable dollar-neutral factor-mimicking portfolio for each firm type m.

Constraints on factor construction (enforced per factor, per time). To ensure each factor is implementable and not dominated by microcaps or illiquid names, we enforce:

$$\begin{aligned}
&\mathbf{1}^\top w^{(m)}_{t-1} = 0, \qquad \|w^{(m)}_{t-1}\|_1 \le L_f, \qquad \|w^{(m)}_{t-1}\|_\infty \le w^{(f)}_{\max},\\
&w^{(m)}_{t-1} \in \mathcal{W}_{t-1}, \qquad A_{t-1} w^{(m)}_{t-1} = 0 \;\; \text{(optional: market/sector neutrality for factor purity).}
\end{aligned} \tag{15}$$

If (14) violates bounds (e.g., $\|w\|_\infty$), we apply clipping and re-normalize while preserving self-financing (details in implementation notes).

Firm-type factor returns. The corresponding factor return is

$$f_{m,t} = \sum_{i=1}^{N} w^{(m)}_{i,t-1}\, r_{i,t}. \tag{16}$$

Stacking factors gives $f_t = (f_{1,t},\ldots,f_{M,t})^\top \in \mathbb{R}^M$.

Time-series factor model and estimated exposures. Asset pricing tests are conducted using time-series regressions:

$$r_{i,t} = \alpha_i + \beta_i^\top f_t + \varepsilon_{i,t}, \qquad \beta_i = (\beta_{i,1},\ldots,\beta_{i,M})^\top, \tag{17}$$

where $\beta_{i,m}$ denotes exposure of asset i to firm-type factor m. Model validity is assessed via tests of $\alpha_i = 0$ and by examining mean returns and Sharpe ratios of $f_{m,t}$.

From factor returns to a tradable strategy (factor timing / tangency over firm-type factors). Specification B naturally produces a factor return panel $\{f_t\}$. We form a tradable *factor-combination* portfolio with weights $a_{t-1} \in \mathbb{R}^M$ on the firm-type factors. Define factor expected returns $\hat{\mu}^{(f)}_{t|t-1}$ and factor covariance $\hat{\Sigma}^{(f)}_{t-1}$ estimated using only data up to $t-1$ (rolling or expanding window). We solve the constrained factor-level optimization:

$$\begin{aligned} a^\star_{t-1} \;\in\; \arg\max_{a\in\mathbb{R}^M} \quad & a^\top \hat{\mu}^{(f)}_{t|t-1} \;-\; \frac{\gamma_f}{2}\, a^\top \hat{\Sigma}^{(f)}_{t-1}\, a \;-\; \kappa_f\, \|a - a^\star_{t-2}\|_1 \\ \text{s.t.} \quad & \mathbf{1}^\top a = 0 \qquad \text{(optional: self-financing across factors)} \\ & \|a\|_1 \le L_a \qquad \text{(gross exposure cap across factors)} \\ & \|a\|_\infty \le a_{\max} \qquad \text{(single-factor concentration cap).} \end{aligned} \tag{18}$$

Implied single-name holdings. The induced single-name portfolio weights are

$$w^{(\text{impl})}_{t-1} = \sum_{m=1}^{M} a^\star_{m,t-1}\, w^{(m)}_{t-1}, \tag{19}$$

which inherit self-financing and (optionally) neutrality properties from (15) and (18).

Final return output (Specification B). The realized return of the factor-combination strategy over $[t-1, t]$ is equivalently

$$R^{(B)}_t = (a^\star_{t-1})^\top f_t = \sum_{i=1}^{N} w^{(\text{impl})}_{i,t-1}\, r_{i,t} \;-\; \mathrm{TC}^{(f)}_t, \tag{20}$$

where $\mathrm{TC}^{(f)}_t$ denotes the transaction cost model applied at the factor-combination (and implied single-name) level, and all constraints in (15)–(18) are enforced each period.

2.5 Validation

For a start in model validation, we can use this python wrapper that these other ucla dudes made where you essentially just enter in forcasted residual price and a signal (i.e. stat arb residual indicator), and it outputs a 100 ish page pdf of all different portfolio weights and their respective PnLs:

`https://github.com/mcucurin/alphamark/tree/main`

The readme doesn't say anything reference the AlphaMark_Guide.pdf file in the repo for more details.

References

Eugene F Fama and Kenneth R French. Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1):3–56, 1993.

Eugene F Fama and Kenneth R French. A five-factor asset pricing model. *Journal of Financial Economics*, 116(1):1–22, 2015.

Bryan T Kelly, Seth Pruitt, and Yinan Su. Characteristics are covariances: A unified model of risk and return. *Journal of Financial Economics*, 134(3):501–524, 2019.